**111C0061 - Spring & Summer 2025**

# Signals and Systems

Tuesday 8:00am-9:35am, Thursday 10:00am-12:25pm, 紫金港西 2-310 (多)

## Instructors


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## TAs


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## Course Website

学在浙大 http://course.zju.edu.cn/

## Course Objectives

Learning mathematical methods used to describe signals and to analyze and synthesize systems.

## Prerequisites

Calculus, Linear Algebra, Principles of Electronic Circuits

## Required Textbook

Oppenheim, Willsky, and Nawab, Signals and Systems, 2nd Edition, 电子工业出版社, 2002 (OWN)

## References

1. Buck, Daniel, and Singer, Computer Explorations in Signals and Systems Using Matlab, Second Edition, Prentice Hall, 2002 (BDS)

2. 张昱, 信号与系统实验教程, 人民邮电出版社, 2005

## Grading

Grades are based on the following weights:

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| Homework + Project | 10% + 20% |
| Midterm Exam | 30% |
| Final Exam | 40% |

## Problem Sets

For the most part, problem sets will be assigned on a **weekly** basis. Each will be made available on the web on the day of a lecture and collected a week after. A detailed schedule of dates is included in the course calendar. **Problem sets that are handed in late will be acknowledged, but a grade multiplied by 0.5 will be recorded.**

These assignments are intended to provide you with practice using the ideas and concepts we will be developing and are an essential part of the process of mastering the course. Completing them is important to you in learning the material. Grading them is important to us in assessing how you are doing.

Solutions to each problem set will generally be made available on the day of the first lecture following the problem set due date.

## Projects

During the semester there will be **four MATLAB-based assignments** that are intended to complement the problem sets by providing you with a different experience in mastering and gaining an appreciation for the course material. The dates on which these will be handed out or posted on the web and the dates when they are due are listed in the course calendar. The write-ups for the labs should be fairly concise, including graphs, Matlab code, brief descriptions of what you did, etc. **As with problem sets, if a lab is handed in late, it will be acknowledged but the grade will be multiplied by 0.5.**

## Cooperative Work

We fully recognize the potential value of students working or studying together, and we do not have any objection to this kind of cooperation, **so long as all participants are involved in all aspects of the work**, not with each doing only a fraction of the assignment. In particular, when you hand in a paper with your name on it, we assume that you are certifying that the details presented are entirely your own work and that you played a substantial role at the conception stage. If part of the work was done cooperatively, that part should be indicated and the names of the students involved in the collaboration should be listed on each

submitted paper.

**Exams**

There will be a mid-term exam and a final exam during the Final Examination Period of the Spring and Summer semesters, respectively. Students will have two hours to complete each exam. The final exam will cover all of the material in the subject, but may be weighted more heavily on material not covered in the mid-term exam. All exams are closed book, but you will be allowed one A4 sheet of notes (both sides) for the mid-term exam and two A4 sheets of notes (both sides) for the final exam. Calculators are not allowed.

**Tentative Outline**

| Num | SES# | DATES | TOPICS | ACTIONS | READINGS |
|---|---|---|---|---|---|
| 1 | L1 | Feb. 18 | Signals and Systems (Z) | | Ch 1.0-1.6 |
| 2 | L2 | Feb. 20 | Discrete-time (DT) & Continuous -time (CT) Systems (Z) | | Supplementary notes Ch. 1-2 |
| 3 | L3 | Feb. 25 | Feedback, Cycles, and Modes (J) | | Supplementary notes Ch. 3-5 |
| 4 | Lab1 | Feb. 27 | Lab: Matlab (C, TA) | Project 1 assignment PS 1 Due | |
| 5 | R1 | Mar. 4 | Recitation (J, TA) | | |
| 6 | L4 | Mar. 6 | Laplace Transform (J) | | Ch. 9.1-9.10 |
| 7 | L5 | Mar. 11 | Z Transform, Relations between CT and DT (J) | | Ch. 9.6-9.9, 10 |
| 8 | Lab2 | Mar. 13 | Lab: LTI + Feedback (C, TA) | Project 2 assignment PS 2 Due | |
| 9 | R2 | Mar. 18 | Recitation (J, TA) | | |
| 10 | L6 | Mar. 20 | Convolution (J) | | Ch. 2 |
| 11 | L7 | Mar. 25 | Frequency Response + Bode Plots (J) | | Ch. 9.4, 10.4, 3.8-3.11, 6.0-6.2, 6.5 |
| 12 | Lab3 | Mar. 27 | Lab: Laplace, Z-Transform (C, TA) | Project 3 assignment PS 3 Due | |
| 13 | R3 | Apr. 1 | Recitation (J, TA) | | |
| 14 | L8 | Apr. 3 | DT Feedback and Control (J) | | Ch. 11.0-11.2 |

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| 15 | L9 | Apr. 8 | CT Feedback and Control (J) | | Ch. 11.0-11.2 |
| 16 | R4 | Apr. 10 | Recitation (J, TA) | PS 4 Due | |
| 17 | L10 | Apr. 15 | CT Fourier Series (Z) | L10 and E1 may exchange | Ch. 1.3, 3 |
| 18 | E1 | Apr. 17 | Mid term Exam | | |
| 19 | L11 | Apr. 22 | CT Fourier Series (cont'd) (Z) | | Ch. 1.3, 3 |
| 20 | L12 | Apr. 24 | CT Fourier Transform (Z) | | Ch. 4 |
| 21 | L13 | Apr. 29 | CT Fourier Transform (cont'd) (Z) | | Ch. 4 |
| 22 | L14 | May 6 | DT Fourier Series (Z) | PS 5 Due | Ch. 3.7-3.11 |
| 23 | L15 | May 8 | DT Fourier Transform & Fourier Relations (Z) | | Ch. 5 |
| 24 | R5 | May 13 | Recitation (J, TA) | PS 6 Due | |
| 25 | Lab4 | May 15 | Lab: Fourier Series & Transform (C, TA) | Project 4 assignment | |
| 26 | L16 | May 20 | Sampling (Z) | | Ch. 7.1-7.3 |
| 27 | L17 | May 22 | Sampling (cont'd) (Z) | | Ch. 7 |
| 28 | L18 | May 27 | Modulation (Z) | PS 7 Due | Ch. 8.0-8.3 |
| 29 | L19 | May 29 | Modulation (cont'd) (Z) | | Ch. 8.4-8.8 |
| 30 | R6 | Jun. 3 | Recitation (Z, TA) | PS 8 Due | |
| 31 | Lab5 | Jun. 5 | Lab: Sampling (C, TA) | | |
| | E2 | TBD | Final Exam | | |